Panum Crater

Dome sequence, degassing textures and a hybrid effusive–explosive eruption at the north end of the Mono–Inyo volcanic chain

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EPS 118 field report
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What the mapping set out to answer

1. In what order did the domes erupt?
2. What textures does each dome preserve?
3. How efficiently did the magma degas on ascent?
4. Was the eruption effusive, explosive or both?

Setting

Panum sits on the Walker Lane, where dextral Pacific–North America motion meets Basin and Range extension. Long Valley Caldera, on that boundary, has about 3 million years of mafic to dacitic volcanism; E–W extension and subsidence there let N–S dikes rise and are thought to have built the Mono–Inyo chain about 50–35 ka. Panum formed in the North Mono eruption about 600 years ago, fed by a N–S dike beneath the northern 6 km of the chain. Magma met groundwater-saturated alluvium, driving hydrovolcanic, sub-Plinian to Plinian activity.

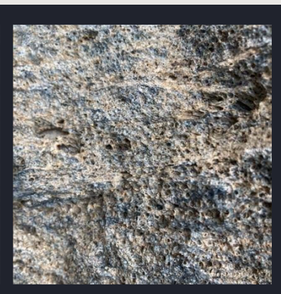
Eruption sequence

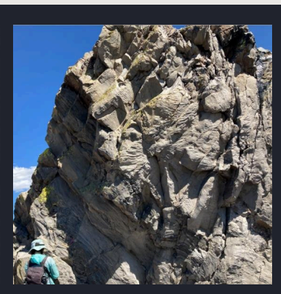
1. **Phreatic blast.** Deposits the North Mono airfall tephra.
2. **Throat clearing.** Massive unsorted breccia, lapilli, accidental sediment and Sierran clasts.
3. **Pyroclastic flows and surges.** Lapilli gravels, laminated pumice and cross-bedded ash with block bombs, common on the SW dome; the triangular facet on the west tephra ring dates from here.
4. **Phreatic explosion.** Blasts out glassy rhyolite breccia and breadcrust pumice, collapses the inner crater and sends the Panum block-and-ash flow N–NW.
5. **Pyroclastic flow.** A layer of ash and lapilli.
6. **Final extrusion.** Flow-banded obsidian, pumice and stony rhyolite build the North Dome and its tephra ring.

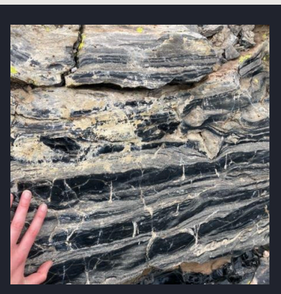
What drove it

The North Mono dike is thought to have intruded beneath the northern 6 km of the chain as the lithosphere rebounded elastically during crustal extension. Feeder dikes here are narrow, under 3 m, so magma mixing together with tectonic strain probably over-pressurized the Mono Craters reservoir.


A Stony rhyolite

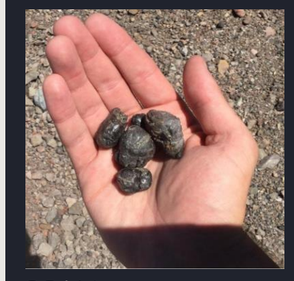

B Vesicular rhyolite + obsidian


C Backfill breccia

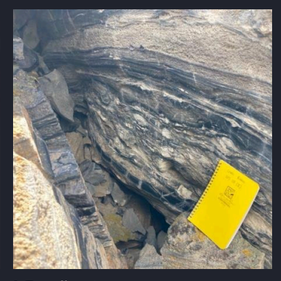

D Obsidian


E Breadcrust


F Flow banding


G Pele's tears

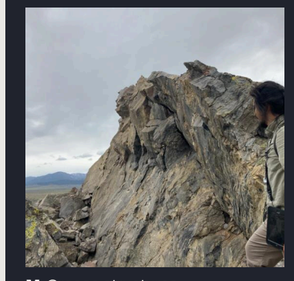

H Devitrification

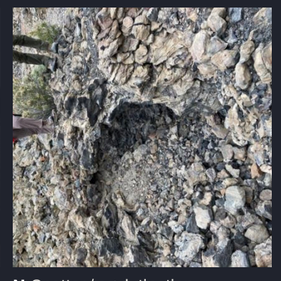

I Boudinage

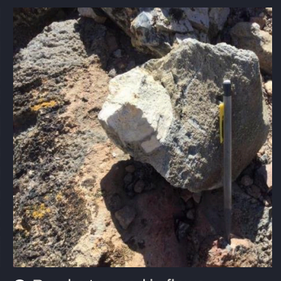

J Auto-brecciation

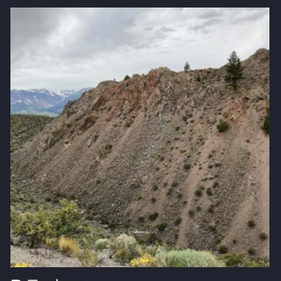

K Fumarolic alteration


L Tuffisite


M Crease structures


N Spatter / agglutination


O Bombs trapped in flows


P Tephra cone


Q En echelon shearing

Field textures

Letters are cited in the text below.

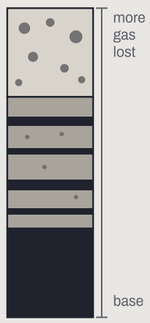
Four domes, oldest to youngest

Southwest	East	South	North
Cream stony rhyolite **A**; fragment of an earlier dome	Stony grading up to pumiceous rhyolite **B**	N–S graben; cut by the North Dome fissure	Two-thirds of the surface; pumice and planar obsidian

older → younger

The SW and East domes are shattered and pushed to the crater flanks. More pumice in the East Dome hints it is a little younger than the SW Dome. The North Dome diapir split the South Dome along its fissure, stretching the flow banding into boudins **I**.

Inside a dome



- **Top:** pumiceous rhyolite, most vesicular.
- **Middle:** microvesicular stony rhyolite interbedded with obsidian; vesicles grow upward.
- **Base:** massive obsidian.
- The South Dome adds a 10 m sheath of foliated pumice wrapped in a 1 m obsidian shell over its stony rhyolite and banded obsidian **D**.
- Obsidian proportions vary a lot from place to place.

Along the North Dome fissure

Near-vertical, outward-dipping flow-banded stony rhyolite with breadcrust rinds **E** forms crease structures **M** parallel to the N–S fissure. Backfill breccia **C** fills the central fissure and spines, and the fissure walls are heavily auto-brecciated **J** by magma injection and gas expansion. Tuffisite veins run parallel to the fissure at its south end and on both sides of the South Dome graben. Talus on the north side coarsens from cobbles to boulders, pointing to a large pyroclastic blast to the north.

A record of degassing

Vesicle size tracks how much gas escaped and where a layer sits in the dome. Fast depressurization froze obsidian blocks in the South Dome graben and fissure, some devitrified **H**. Red-orange fumarolic staining **K** marks intense vent degassing, where hot water vapour oxidized iron in the rock; fumaroles line the fissures and dome flanks. Breadcrust rinds **E** and Pele's tears **G**, small blebs of quenched melt from the tephra band around the East Dome, record gas escaping as melt cooled.

Effusive and explosive at once

Ejecta thrown steeper than 45° fell back into the fissure and was welded into new lava as backfill breccia **C**. Spines cracked as they extruded and ash blew into the cracks as tuffisite **L**. Bombs sit trapped in flows **O**. Spatter cones and vents just east of the South Dome fissure **N** sit beside small tephra cones holding Pele's tears, and the tephra ring around the East Dome and vesicular stony rhyolite under the western tephra ring point to steady ash venting during effusion.

Timing

The whole North Mono sequence, from the first phreatic blast to the North Dome, left no lake or wind-blown beds between its pyroclastic layers, so it likely ran its course in a few months. The triangular facet on the west side of the tephra ring belongs to the early pyroclastic phase and was later buried by North Dome tephra. Phreatic blasts and small effusive eruptions of thick, silica-rich rhyolite repeatedly went together.

Black, Manga & Andrews (2016) Bull. Volcanol. 78, 57.
Reches & Fink (1988) JGR 93(B5), 4321–4334.
Sieh & Bursik (1986) JGR 91(B12), 12539–12571.
Unruh, Humphrey & Barron (2003) Geology 31, 327–330.
Wohletz (1987) Pyroclasts from hydrovolcanic eruption sequences.